F79PS Further Statistical Methods Assessed Project 2025

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Course: Statistical Data Science

Statistical terminology

All statistical tests and calculations were done using SPSS, below are clarification on the definitions used in this report:

1. For all output obtained from SPSS, "Sig" refers to the P-Value, that is the probability of obtaining the observed data, assuming the null hypothesis of a statistical test is true.
2. For all output obtained from SPSS, "df" refers to degree of freedom, that is number of independent values that are free to change in a dataset after the estimation of a parameter.
3. In a statistical test, H_0 is the notation for the null hypothesis and H_1 is the notation for the alternative hypothesis. We will assume the null hypothesis is true unless there is enough evidence to reject, then we will favor the alternative hypothesis.

Question 1

Introduction

This report discusses a two-way ANOVA to investigate the effects of gender (male, female) and age group (<37, 37-49, 50+) an individual's overall sleepiness level. The study utilised the Total Sleepiness and Associated Sensations Scale (TotSAS).

Hypotheses

Listed below are the three main hypotheses we focus on testing.

1. Main Effect of Gender:

$$H_0: The\ mean\ TotSAS\ score\ between\ male\ and\ female\ is\ the\ same.$$

$$H_1: The\ mean\ TotSAS\ score\ between\ male\ and\ female\ is\ not\ the\ same.$$

2. Main Effect of Age group:

$$H_0: The\ mean\ TotSAS\ score\ is\ the\ same\ across\ all\ age\ groups\ (< 37, 37-49, 50+).$$

$$H_1: At\ least\ one\ of\ the\ mean\ TotSAS\ score\ for\ one\ of\ the\ age\ group\ is\ different.$$

3.Interaction Effect (Gender*Age group)

$$H_0: There\ is\ interaction\ between\ gender\ and\ age\ group\ on\ the\ TotSAS\ .$$

$$H_1: There\ is\ interaction\ between\ gender\ and\ age\ group\ on\ the\ TotSAS\ scores.$$

Assumptions

Before carrying out the two-way ANOVA, three assumptions must be fulfilled.

Assumption 1: Independence of observations: Independence arises from the way the experiment is set up. This assumption is fulfilled as the individuals were selected randomly.

Assumption 2: Normality. To check whether the data is normally distributed, the dot plots for both male and female across all age groups are shown in Figure 1.1.

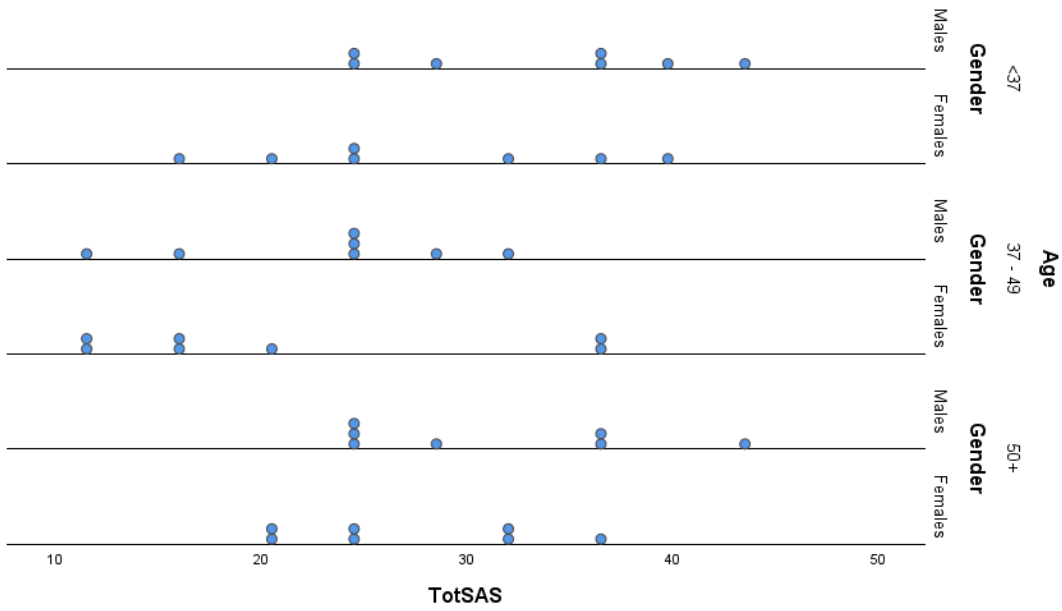

Figure 1.1 Dot plot of the TotSAS scores for each combination of gender and age group.

The dot plot seems reasonably normal, but it is hard to make a conclusion due to the small sample size for each age group. To formally check for normality, the Kolmogorov-Smirnov (K-S) test and Shapiro-Wilk (S-W) test is carried out.

For both the K-S test and S-W test, the hypotheses are:

$$H_0: The\ data\ follows\ a\ normal\ distribution.$$

Tests of Normality

Gender	Age		Kolmogorov-Smirnov[a] Statistic	df	Sig.	Shapiro-Wilk Statistic	df	Sig.
Males	<37	TotSAS	.191	7	.200*	.909	7	.392
	37 - 49	TotSAS	.198	7	.200*	.972	7	.913
	50+	TotSAS	.236	7	.200*	.896	7	.305
Females	<37	TotSAS	.140	7	.200*	.955	7	.776
	37 - 49	TotSAS	.255	7	.186	.861	7	.156
	50+	TotSAS	.257	7	.180	.898	7	.318

*. This is a lower bound of the true significance.

a. Lilliefors Significance Correction

$$H_1: The\ data\ does\ not\ follow\ a\ normal\ distrbution.$$

Table 1.2 Results for Kolmogorov-Smirnov test and Shapiro-Wilk test for each combination

The cells in "green" are the P-values for the Kolmogorov-Smirnov test and the cells in "yellow" are the P-values for the Shapiro-Wilk test. The results for both normality tests show that there is no eivdence agaisnt normality, hence the normality assumption is valid. It is important to note that the Shapiro-Wilk test is more suitable as the sample size for each combination is small, that is n = 7.

Assumption 3: Equality of variances

To check for equality of variances, Levene's test is carried out and the results are shown in Table 1.2.

Let $\sigma^2_{i,j}$ be the population variance of the TotSAS scores for all combinations of gender and age groups where i = (male, female) and j = (<37, 37-49, 50+).

$$H_0: \sigma^2_{male,<37} = \sigma^2_{male,37-49} = \sigma^2_{male,50+} =$$

$$\sigma^2_{female,<37} = \sigma^2_{female,37-49} = \sigma^2_{female,50+}$$

H_1: At least one of the population variance is different from the others

Levene's Test of Equality of Error Variances[a,b]

		Levene Statistic	df1	df2	Sig.
TotSAS	Based on Mean	.844	5	36	.527
	Based on Median	.333	5	36	.890
	Based on Median and with adjusted df	.333	5	23.274	.888
	Based on trimmed mean	.815	5	36	.547

Tests the null hypothesis that the error variance of the dependent variable is equal across groups.

a. Dependent variable: TotSAS

b. Design: Intercept + Gender + Age + Gender * Age

Table 1.3 Results for Levene's Test of equality of error variances

From the results of the Levene's Test, the test statistic is 0.844 and the P-value is 0.527, suggesting that there is no evidence to reject the null hypothesis that the population variances for all the combinations of gender and age groups are equal. Hence, equality of variances is assumed.

Here concludes the validity checks for all three assumptions of independence, normality and equality of variances. All three assumptions are fulfilled, hence the two-way ANOVA is suitable to be carried out.

Two-way Analysis of Variances (ANOVA)

A two-way ANOVA was carried out in SPSS and Table 1.4 shows a summary of the results.

Tests of Between-Subjects Effects

Dependent Variable: TotSAS

Source	Type III Sum of Squares	df	Mean Square	F	Sig.	Partial Eta Squared	Noncent. Parameter	Observed Power[b]
Corrected Model	658.762[a]	5	131.752	1.956	.109	.214	9.780	.590
Intercept	30834.381	1	30834.381	457.774	<.001	.927	457.774	1.000
Gender	144.857	1	144.857	2.151	.151	.056	2.151	.298
Age	494.905	2	247.452	3.674	.035	.170	7.347	.639
Gender * Age	19.000	2	9.500	.141	.869	.008	.282	.070
Error	2424.857	36	67.357					
Total	33918.000	42						
Corrected Total	3083.619	41						

a. R Squared = .214 (Adjusted R Squared = .104)

b. Computed using alpha = .05

Table 1.4 Results of two-way ANOVA

To check whether there is interaction between factors gender and age group, an Interaction Test is performed.

Interaction Test

H_0: There is no interaction between gender and age group on TotSAS scores.

H_1 : There is interaction between gender and age group on TotSAS scores.

From Table 1.4, the P-value for the interaction test is 0.869 (Highlight in "green"), this suggests that there is no evidence to reject the null hypothesis that there is no interaction between gender and age group on TotSAS scores. Therefore, different combinations of gender and age groups will not affect the TotSAS scores.

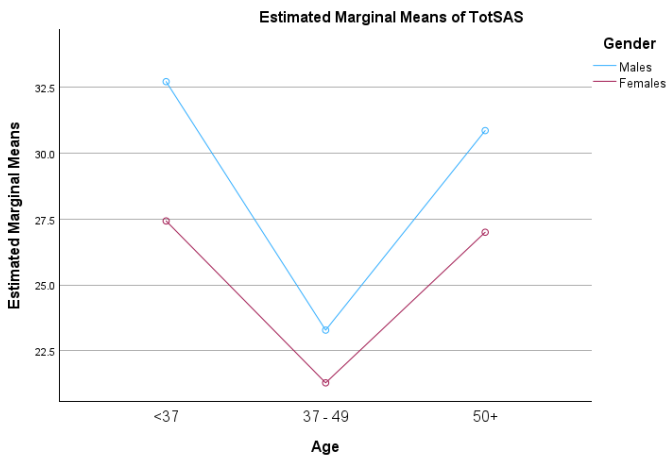

Figure 1.5(a) (left) visualizes the estimated marginal means of TotSAS scores for age groups and Figure 1.5(b) (right) visualizes the estimated marginal means of TotSAS scores for gender.

From the graphs above, it is reasonable to say that there is no significant interaction between gender and age as the lines for both graphs are reasonably parallel, that is they do not differ significantly enough to contradict the conclusion previously obtained from the interaction test.

Since the interaction is not significant, tests for gender and age group will be carried out seperately.

Test for Gender

The hypotheses for the statistical test for Gender are,

$$H_0: There\ is\ no\ difference\ in\ TotSAS\ scores\ between\ male\ and\ female.$$

$$H_1: There\ is\ a\ difference\ in\ TotSAS\ scores\ between\ male\ and\ female.$$

From Table 1.4 which shows the results of the two-way ANOVA, the P-value is 0.151 (highlighted in "yellow"), which suggests there is no evidence to reject the null hypothesis that there is no difference on TotSAS scores due to different genders. Hence, we can say that difference in gender will not affect the TotSAS scores.

Test for Age Group

The hypotheses for the Age Test are,

$$H_0: There\ is\ no\ difference\ in\ TotSAS\ scores\ among\ age\ groups\ < 37, 37-49, and\ 50+.$$

$$H_1: There\ is\ a\ difference\ in\ TotSAS\ scores\ among\ age\ groups\ < 37, 37-49, and\ 50+.$$

From Table 1.4 which shows the results for the two-way ANOVA, the P-value is 0.035 (highlighted in "lightblue"). This suggests that there is some evidence to reject the null hypothesis that there is a difference in TotSAS scores due to different age groups. Hence, we can say that difference in age groups has an effect on the TotSAS scores.

Follow-up for Age Group Test

2. Age

Dependent Variable: TotSAS

Age	Mean	Std. Error	95% Confidence Interval	
			Lower Bound	Upper Bound
<37	30.071	2.193	25.623	34.520
37 - 49	22.286	2.193	17.837	26.734
50+	28.929	2.193	24.480	33.377

Table 1.6 Means and standard error of TotSAS scores for different age groups

Based on Table 1.6, the mean TotSAS score for each age group is highlighted in "green". Age group <37 has the highest mean score at 30.071, this is followed by age group 50+ then age group 37-49. This suggests that individuals younger than 37 generally feel a higher level of sleepiness than individuals aged 37 and older. The results also suggests that individuals aged 50 and above generally feel a higher level of sleepiness than those in age group 37-49.

Furthermore, the confidence intervals of mean TotSAS scores for all three age groups (highlighted in "pink") are slightly overlapping with each other, with the exception of age groups <37 and 50+ which overlaps quite a bit. However, overall the overlapping confidence intervals supports the conclusion that there is no significant difference in TotSAS scores among different age groups.

Value of power and eta-squared for Age Group Test

From the results in Table 1.4, the power of the Age Test is 63.9% (highlighted in "pink"), which indicates that the test is quite powerful. Table 1.4 also shows that the value of eta-squared for the Age Test is 0.170 (highlighted in "red"), which indicates that age explains 17% of the variability in the TotSAS scores.

Conclusion

In conclusion, since there is no significant difference in TotSAS scores between male and female and the mean TotSAS score in age group <37, we conclude that the mean TotSAS scores is highest individuals younger than 37 and lowest for those in age group 37-49, suggesting that individuals below the age of 37 experience the highest levels of sleepiness overall and individuals within 37-49 years old experience the lowest levels of sleepiness overall.

Question 2

Introduction

The aim of Question 2 is to analyze a dataset containing the physical measurements (sex, length, diameter, height, whole weight, shucked weight, viscera weight) of 400 abalones using Principal Compenents Analysis (PCA) and Factor Analysis (FA) in order to determine which factors could predict the age of an abalone.

Principal Components Analysis (PCA) and Factor Analysis (FA) are both used to simplify data by reducing the number of components and extracting meaningful insights. PCA aims to reduce the number of dimensions while maximizing the total variance captured. FA aims to identify underlying latent variables and find physical interpretations of new components or factors by grouping the original components if possible.

Kaiser-Meyer-Olkin (KMO) test and Barlett's Test of Sphericity

Firstly, the data is assessed using the KMO test and Barlett's Test to check whether it is suitable for PCA and FA.

If the variables are correlated, then the data is suitable for PCA or FA.

KMO and Bartlett's Test

Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		.880
Bartlett's Test of Sphericity	Approx. Chi-Square	5280.667
	df	15
	Sig.	<.001

Table 2.1 KMO and Berlett's Test Results

The Kaiser-Meyer-Olkin Measure of Sampling Adequacy measures the proportion of variance in each variable that might be caused by underlying factors. From Table 2.1, the value for the KMO test is 0.880 (highlighted in "green"), which is quite high (close to 1). This suggests that there is a high degree of common variance hence PCA or FA may be useful for the data.

Barlett's test of sphericity checks whether the variables are unrelated by testing if the correlation matrix is significantly different from an identity matrix.

The hypotheses for Barlett's test is,

$$H_0: \text{There is no correlation between variables.}$$

$$H_1: \text{There is correlation between variables.}$$

From Table 2.1, the P-value for Barlett's Test is smaller than 0.001 (highlighted in "yellow"). Hence, there is overhwelming evidence to reject the null hypothesis which indicates that there is correlation between the variables. Therefore the data is suitable for PCA or FA.

Both tests gives to same conclusion that is there is correlation between the variables in the dataset hence PCA or FA is suitable to be performed.

Descriptive Statistics

Table 2.2 below shows the frequencies of male, female, and infant abalones in the dataset.

Sex

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	F	153	38.3	38.3	38.3
	I	85	21.3	21.3	59.5
	M	162	40.5	40.5	100.0
	Total	400	100.0	100.0	

Table 2.2 Frequency table for male, female, and infant

From Table 2.2, the dataset contains a total of 315 adult abalones (162 males + 153 females) and 85 infants.

Descriptive Statistics

	N	Minimum	Maximum	Mean	Std. Deviation
Length	400	.17	.73	.5255	.11864
Diameter	400	.13	.60	.4108	.09841
Height	400	.00	.24	.1397	.03966
Whole_weight	400	.03	2.55	.8533	.50057
Shucked_weight	400	.01	1.13	.3582	.21533
Viscera_weight	400	.01	.54	.1871	.10943
Valid N (listwise)	400				

Table 2.3 Descriptive Statistics for physical measurement of abalone

From Table 2.3, the standard deviations do not differ greatly with the largest at 0.50057 and the smallest at 0.03966. Hence, the variances of the variables will also not differ greatly.

Correlation matrix

Correlation Matrix

		Length	Diameter	Height	Whole_weight	Shucked_weight	Viscera_weight
Correlation	Length	1.000	.989	.915	.925	.900	.909
	Diameter	.989	1.000	.925	.928	.899	.908
	Height	.915	.925	1.000	.903	.864	.892
	Whole_weight	.925	.928	.903	1.000	.970	.959
	Shucked_weight	.900	.899	.864	.970	1.000	.935
	Viscera_weight	.909	.908	.892	.959	.935	1.000

Table 2.4 Correlation matrix for physical measurements of abalone

From the correlation matrix above, all the variables are strongly positively correlated (all values more than 0.8), the conclusion is the same as previously obtained in the KMO and Barlett's Test, which is that PCA and FA is highly appropriate for this dataset.

Note that since most of the variables are continuous variables, Pearson's Correlation test is used. Only the variable "Sex" is not included as it is a categorical variable.

Principal Components Analysis (PCA)

Now that is it confirmed the variables are highly correlated, Principal Components Analysis can be carried out. The results are shown below.

Total Variance Explained

	Initial Eigenvalues			Extraction Sums of Squared Loadings		
Component	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	5.608	93.471	93.471	5.608	93.471	93.471
2	.191	3.191	96.662	.191	3.191	96.662
3	.105	1.753	98.416			
4	.062	1.032	99.447			
5	.023	.385	99.832			
6	.010	.168	100.000			

Extraction Method: Principal Component Analysis.

Table 2.5 Results for Principal Component Analysis

The maximum number of components that can be extracted is six, as there are six variables to be considered.

Three criterias are used to determine the number of principal components to extract.

i. Kaiser's criterion – extract components with eigenvalues greater than 1
ii. High total variance explained
iii. Scree plot levels off – plots the eigenvalues against the component numbers

From Table 2.5, there is one eigenvalue over 1, namely component 1, and it accounts for 93.471% (highlighted in "lightblue") of the variance, which is extremely good.

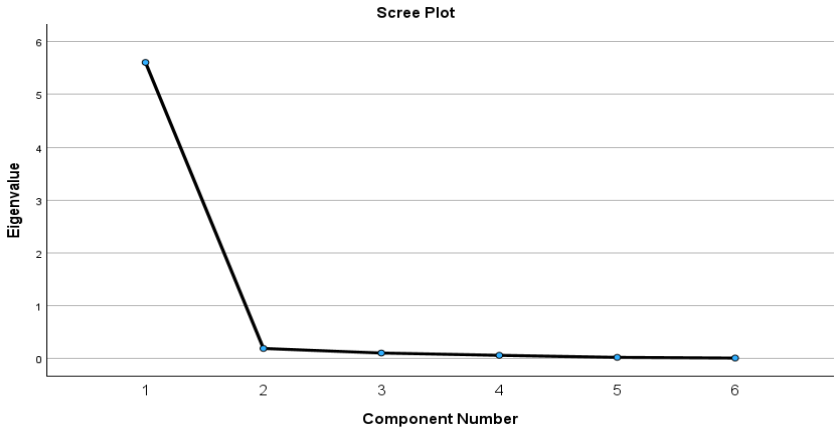

Figure 2.6 Scree Plot for PCA

The scree plot above shows that there is an 'elbow' at the component 2, after which the line starts to flatten out. This indicates that after component 1, subsequent components contribute little explanotory power. However, there is a small drop from components 2 to 3, which suggests that component 2 still contributes to some variance, even though the eigenvalue is quite small, that is 0.191, it still explains a small amount of the total variance, that is 3.191% (highlighted in "yellow"). Hence, it is reasonable to extract two components, that is components 1 and 2, as together they explain for at 96.66% of the total variance. This is quite good for six variables. Extracting components 1 and 2 also allows us to do rotation in Factor Analysis which can then be used to compare with our results obtained from PCA.

Communalities

Communalities represent the proportion of each variable's variance that is explained by the extracted factors.

A value close to 1 indicates high communality and a value close to 0 indicates low communality. Note that PCA uses 1 as the initial communality for all variables.

Communalities

	Initial	Extraction
Length	1.000	.972
Diameter	1.000	.979
Height	1.000	.935
Whole_weight	1.000	.983
Shucked_weight	1.000	.974
Viscera_weight	1.000	.956

Extraction Method: Principal Component Analysis.

Table 2.7 Table of communalities

The communalities table above shows quite high values ranging from 93.5% to 98.3%. This suggests that the two components extracted is excellent at explaining the variance in each of the six variables.

Component Matrix

The component matrix shows the eigenvectors (loadings) and the associated eigenvalues extracted. Loadings represent the correlation between each variable and the associated extracted comonent.

Component Matrix[a]

	Component	
	1	2
Length	.972	.164
Diameter	.974	.175
Height	.948	.192
Whole_weight	.980	-.149
Shucked_weight	.960	-.229
Viscera_weight	.966	-.150

Extraction Method: Principal Component Analysis.

a. 2 components extracted.

Table 2.8 Component Matrix

The linear combinations of standaridised variables for the first principal component is:

$$Y_1 = 0.972\, Z_1 + 0.974\ Z_2 + 0.948\ Z_3 + 0.980\ Z_4 + 0.960\ Z_5 + 0.966\ Z_6$$

where Z represents the standardised version of the variables (due to z-score transformation).

Standardisation is to ensure that variables with larger scales will not dominate over the variables with smaller scales during the analysis.

For principal component 1, the loadings for all six variables are positive and quite high (all values are greater than 0.9). This indicates that principal component 1 is a good representation of the data as it explains most of the variance and represents overall body mass. This means that when one variable is large, all the other variables tend to be large too. This is also supported by the reasoning that all the variables are correlated.

The linear combinations of standaridised variables for the second principal component is:

$$Y_2 = 0.164\, Z_1 + 0.175\, Z_2 + 0.192\, Z_3 - 0.149\, Z_4 - 0.229\, Z_5 - 0.150\, Z_6$$

For principal component 2, variables “Length”, “Diameter”, and “Height” have small positive loadings (0.164 to 0.192) whereas all the weight measurements have negative loadings (-0.229 to -0.149). This suggests that principal component 2 represents shell size or the structural dimensions of an abalone. The contrast also explains additional variation, such as abalones that are bigger in size but lighter than expected and abalones that are smaller in size but heavier than expected.

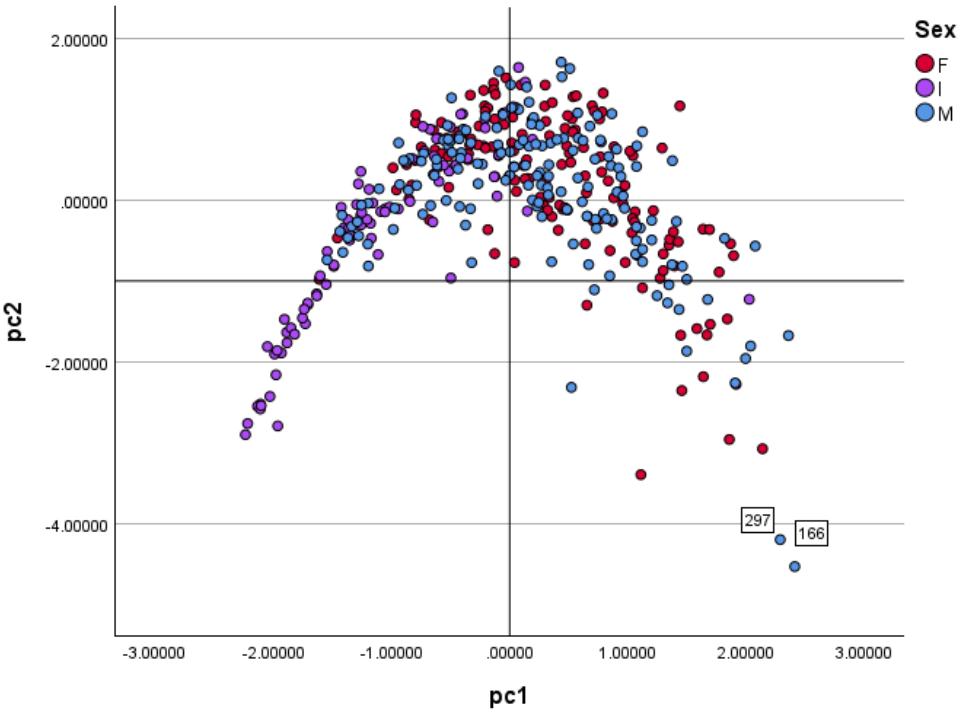

Figure 2.9 Score plot of pc1 vs pc2 by sex

The score plot shows a curved distribution which indicates non-linear growth among the abalones.

Specimens below the horizontal line indicates small shell sizes and specimens behind the vertical line indicates low body mass.

Explainations for specimens in each quadrant is as follows,

Bottom left quadrant: Low body mass and small shell size. Specimens in this quadrant are only infants, this corresponds with their negative scores on factors 1 and 2 which represents overall body size and shell size respectively.

Top left quadrant: Low body mass but large shell, this makes sense as it is the transition period from infant to adult.

Top right quadrant: High body mass and large shell. Specimens in this quadrant are mostly adults and late-stage infants, representing mature and well-developed specimens

Bottom right quadrant: High body mass but small shell. There are a few outliers with high body mass and disporportionately small shell size, such as specimen 297 and 166.

The scores for male and female abalones overlap heavily, with most at the middle of the curve (the middle represents the average body mass), this suggests that both genders have similar physical measurements, so gender does not play a significant role in identifying the age of an abalone based on its physical measurents. Instead, physical measurements can help in indentifying the developmental stage of abalone (i.e. whether it is an infant or an adult).

Factor Analysis

Factor Analysis is used to reduce the number of variables by extracting correlated variables into a smaller set of factors.

Reproduced Correlation Matrix

The reproduced correlation matrix is the correlation matrix based on the extracted factors, that is factors 1 and 2. Residuals shows the differences between the original matrix and the reproduced matrix.

Reproduced Correlations

		Length	Diameter	Height	Whole_weight	Shucked_weight	Viscera_weight
Reproduced Correlation	Length	.972[a]	.976	.953	.928	.896	.915
	Diameter	.976	.979[a]	.957	.929	.895	.915
	Height	.953	.957	.935[a]	.900	.866	.887
	Whole_weight	.928	.929	.900	.983[a]	.975	.969
	Shucked_weight	.896	.895	.866	.975	.974[a]	.962
	Viscera_weight	.915	.915	.887	.969	.962	.956[a]
Residual[b]	Length		.014	-.038	-.004	.005	-.005
	Diameter	.014		-.032	-.001	.004	-.007
	Height	-.038	-.032		.003	-.002	.005
	Whole_weight	-.004	-.001	.003		-.005	-.010
	Shucked_weight	.005	.004	-.002	-.005		-.027
	Viscera_weight	-.005	-.007	.005	-.010	-.027	

Extraction Method: Principal Component Analysis.

a. Reproduced communalities

b. Residuals are computed between observed and reproduced correlations. There are 0 (0.0%) nonredundant residuals with absolute values greater than 0.05.

Table 2.10 Reproduced correlation matrix and residuals

From Table 2.10, the reproduced correlation values are very similar to the original matrix (See Table 2.4), this is further shown as all the residuals are very close to zero. Hence, it is reasonable to say that the two factors extracted gives a good model and provides a good fit to the data.

Rotated Factor Analysis

Rotation maximizes high loadings and minimizes low loadings for each variable, making it easier to interpret the relationships between the variables and factors.

Total Variance Explained

	Initial Eigenvalues			Rotation Sums of Squared Loadings		
Component	Total	% of Variance	Cumulative %	Total	% of Variance	Cumulative %
1	5.608	93.471	93.471	2.925	48.745	48.745
2	.191	3.191	96.662	2.875	47.917	96.662
3	.105	1.753	98.416			
4	.062	1.032	99.447			
5	.023	.385	99.832			
6	.010	.168	100.000			

Extraction Method: Principal Component Analysis.

Table 2.11 Results of Rotated Factor Analysis

After rotation, the proportion of variance explained by factor 1 and factor 2 are quite similar with factor 1 explaining 48.745% (highlighted in "green") and factor explaining 47.717% (highlighted in "pink"). Together, the two factors explain 96.662% of the total variance in the dataset, this is extremely good.

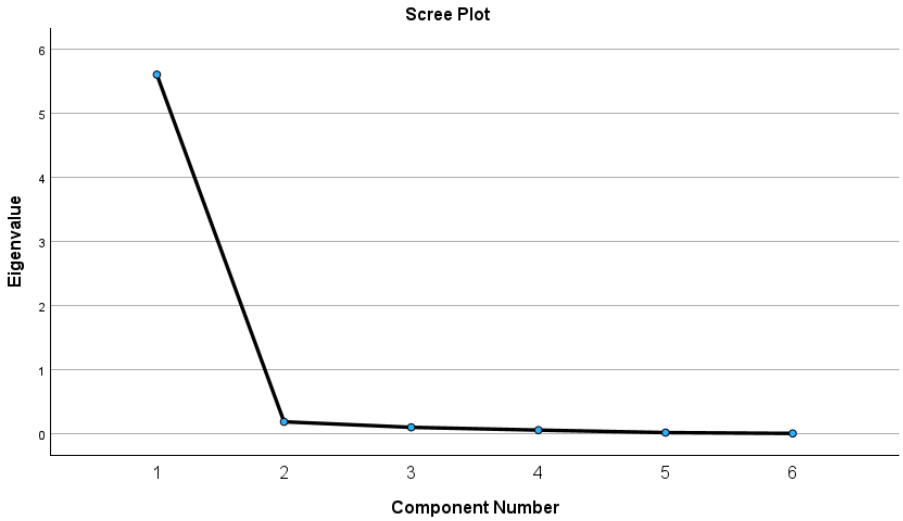

Figure 2.12 Scree plot of rotated factors

It is important to note that rotation does not change the eigenvalues for each component, hence why the scree plot remains the same after rotation. Rotation only redistributes the loadings for each extracted factor but the total variance explained is the same.

Rotated Component Matrix

To provide easier interpretation, orthogonal rotation using Varimax rotation is applied to obtain a rotated component matrix. Varimax rotation was done to maximize the variance of the loadings, this allows for easier interpretation of the loadings in the component matrix.

Rotated Component Matrix[a]

	Component	
	1	2
Length	.806	.568
Diameter	.815	.562
Height	.808	.531
Whole_weight	.591	.796
Shucked_weight	.520	.839
Viscera_weight	.581	.787

Extraction Method: Principal Component Analysis.
Rotation Method: Varimax with Kaiser Normalization.

a. Rotation converged in 3 iterations.

Table 2.13 Rotated Component Matrix

After rotation, factor 1 now represents shell size as the variables "Length", "Diameter", and "Height" have high loadings. Factor 2 now represents body mass, with all the variables related to weight having high loadings.

This makes interpretation easier as each factor is now seen more clearly.

Score Plot after Rotation

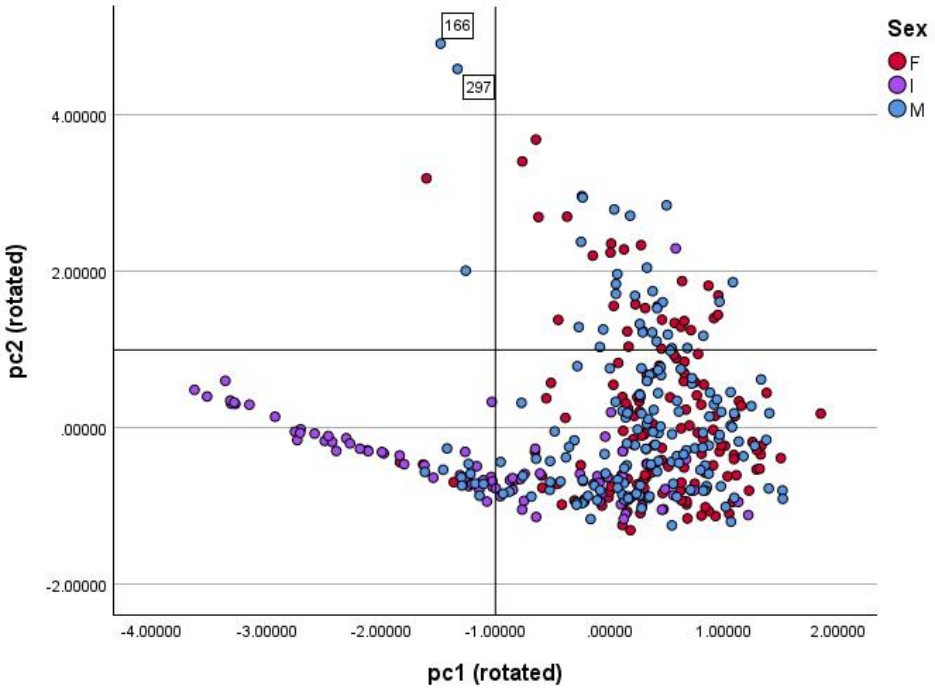

Figure 2.14 Score plot after rotation

There is a clearer seperation between infants and adults. Specimens that are infants mostly lie behind the vertical line and below the horizontal line as it indicates small shell size and low body mass respectively. Adult specimens are mostly clustered in the bottom right quadrant which indicates large shell size and low body mass. This could be due to older specimens as shell growth continues even when soft-tissue growth has slowed. Infant specimens in this quadrant might be in the transition period into an adult abalone. There are also a mix of male and female specimens in the top right quadrant which indicates large shell size and high body mass. Lastly, there are a few outliers, mainly the ones labeled 166 and 297, in the top left quadrant which indicates small shell size and high mass.

Principal Component Analysis Vs Factor Analysis

In this project, Factor Analysis with Varimax rotation is more suitable for data reduction as it provides an easier interpretation of the results by distinguishing between shell size and body mass. Altough both methods succesfully extracted two components, the clear seperation between factors in Factor Analysis provides useful insights when predicting the age of an abalone.

Conclusion

Out of seven descriptive measures, six of the physical measurements (the variable “Sex” was excluded as it is categorical) have successfully been reduced to two that collectively explains 96.662% of the variance from a sample of 400 abalones. The data reduction was done using Principal Factor Analysis and Factor Analysis with Varimax Rotation. The extracted factors are “Body mass” which includes variables “Whole weight”, “Shucked Weight”, and “Viscera Weight” and the second factor is “Shell size” which includes “Length”, “Diameter”, and “Height”. Although these two factors cannot predict the exact age of an abalone, it provides valuable insights on the developmental stage and maturity of an abalone. It is found that infants and younger specimens have lower body mass and smaller shell size whereas adult specimens typically have larger shell size. Specimens with large shells and low body mass may be due to infants transitioning into adults or older specimens.